

# ALEX JIMENEZ

## EDUCATION

**University of California, Merced**  
**Bachelor of Science**, Computer Science And Engineering

## BADGES

- [National Honor Society Member](#)
- [Avid Student Graduate](#)

## Github

- [Unknown903-dev](#)

## PROFESSIONAL SUMMARY

Computer Science and Engineering student focused on full-stack development, cybersecurity, and mobile/cloud-backed applications. Experienced in building React Native and web applications, integrating backend APIs, implementing authentication/session handling, and supporting deployment/testing workflows. Strong background in cybersecurity fundamentals, threat analysis, and secure software design through CodePath, TryHackMe, and project-based development.

## SKILLS

- **Frontend/Mobile:** React, React Native, TypeScript, JavaScript, HTML/CSS, Python, C, C++
- **Backend/Cloud:** Cloudflare Workers, D1, R2, Firebase, FastAPI, Node.js, Flask
- **Authentication/Security:** Google SSO, iOS Keychain, RBAC, password hashing, session handling
- **DevOps/Tools:** Git, GitHub, Xcode, TestFlight, VS Code, Android Studio
- **Testing/QA:** Frontend testing, backend testing, manual QA, debugging
- **Cybersecurity:** Wireshark, Splunk, VirusTotal, Burp Suite, John the Ripper, TryHackMe, Cryptography

## PROJECTS / PROGRAMS

### TIME AND TASK TRACKER RESEARCH PROJECT FOR UC MERCED / CSED ([T4 PROJECT](#))

- Collaborated with UC Merced faculty, students, and CSED on a research-driven project to design a collaborative web application for managing weekly team meeting minutes, tasks, decisions, attendance, and project progress for students in the I2G program.
- Developed and evaluated a full-stack architecture using Angular, FastAPI, and MongoDB, with emphasis on real-time collaboration, version history, structured data collection, and secure access to team and confidential information.
- Researched and prototyped collaborative editing approaches including ShareDB operational transformation (OT), Automerge CRDTs, and MongoDB change streams to evaluate synchronization, conflict resolution, and audit-history strategies.

### PRODUCTION TRACKING MOBILE APP FOR AN INDUSTRY PARTNER

- Led full-stack development for a team-based mobile app supporting employee, manager, and executive workflows for task tracking, QR clock-in, project management, and reporting.
- Integrated React Native frontend screens with Cloudflare Workers APIs, D1 database logic, iOS Keychain session storage, and Firebase push notifications.
- Built and connected features for employee task flows, templates, request approvals/denials, LLM-generated reports, analytical graphs, and shareable PDF report links.
- Supported DevOps and QA through TestFlight deployment preparation, Xcode/iOS debugging, Git workflow organization, and frontend/backend testing.

### WEB HONEYPOT DEPLOYED ON ([UC Merced Interactive Maps Hub](#))

- Achieved enhanced website security through the creation of a web honeypot for Cloudflare environments. Analyzed IP address data and endpoint interactions to identify potential threats. Employed JavaScript to facilitate development and functionality.
- Designed and executed comprehensive bot banning protocols, focusing on threat detection and prevention.
- Evaluated security protocols for SSH configurations on Raspberry Pi systems to enhance overall system integrity.
- Conducted comprehensive review of AWS caching environment variables to ensure optimal configuration.

### CYBERSECURITY STUDENT (CodePath for Cyber 101 / 102)

- Completed hands-on labs in vulnerability assessment, SIEM monitoring, threat detection, and incident response.
- Analyzed network traffic and security scenarios to identify suspicious activity, malicious behavior, and possible attack patterns.
- Collaborated with peers to research emerging threats and develop mitigation strategies for network and system security scenarios.

### MOCK COURSE REGISTRATION DEPLOYED ON ([course registration](#))

- Developed a role-based full-stack web application with a 3-person team using Flask (Python) with SQLAlchemy ORM, simulating a real-world course registration system
- Implemented authentication and authorization with role-based access control (RBAC) for students, instructors, and administrators, enforcing permission boundaries
- Built CRUD functionality for courses, users, enrollments, and grades, enabling students to manage schedules, instructors to assign grades, and admins to manage all system data
- Designed and integrated a dynamic web frontend using JavaScript, HTML, and CSS, connecting to backend Flask routes for seamless user interactions

### MOCK SOCIAL MEDIA PLATFORM

- Engineered a full-stack social media web application utilizing the MEAN stack (MongoDB, Express.js, React, Node.js) to deliver comprehensive Instagram-style functionalities.
- Built RESTful APIs with Node.js and Express.js, supporting full CRUD operations (GET, POST, PUT, DELETE) for users, posts, comments, likes, and follow relationships
- Collaborated in a 3-person team using Git/GitHub, contributing across frontend, backend, debugging, and API integration in an Agile-style workflow
- Implemented secure authentication and authorization, including account creation, login, password hashing and salting, and protected backend routes